olympics

December 7, 2025

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# Project 2: 120 Years of Olympic Games Analysis
# Student: Neil Mahajan
# Course: DAT301
# Data Source: Kaggle - 120 years of Olympic history: athletes and results
# URL: https://www.kaggle.com/datasets/heesoo37/
 ↪120-years-of-olympic-history-athletes-and-results

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import warnings
warnings.filterwarnings('ignore')

# Set style for better-looking plots
plt.style.use('seaborn-v0_8-whitegrid')
sns.set_palette("Set2")

# ====================================================================
# PART 1: BACKGROUND AND PROBLEM DEFINITION
# ====================================================================

print("="*80)
print("OLYMPIC GAMES ANALYSIS: 120 Years of Athletic Excellence (1896-2016)")
print("="*80)

"""
BACKGROUND:
The Olympic Games represent the pinnacle of athletic achievement, bringing␣
 ↪together
athletes from around the world since 1896. This analysis explores 120 years of
Olympic history to understand patterns in medals, athlete characteristics, and
country performance.

DATA SOURCE:
- Dataset: 120 years of Olympic history: athletes and results
- Source: Kaggle (scraped from www.sports-reference.com)
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- URL: https://www.kaggle.com/datasets/heesoo37/
 ↪120-years-of-olympic-history-athletes-and-results
- Records: ~271,116 athlete-event records
- Time Period: 1896 to 2016

RESEARCH QUESTIONS:
1. Which countries have been most successful in Olympic history?
2. How has athlete participation evolved over time (especially for women)?
3. What are the physical characteristics of medal winners vs non-winners?
4. Which sports produce the most medals?
5. Is there a "home advantage" for host countries?
6. How do Summer and Winter Olympics differ in participation and medals?
"""

# ====================================================================
# PART 2: DATA LOADING AND INITIAL EXPLORATION
# ====================================================================

print("\n" + "="*80)
print("LOADING AND EXPLORING DATA")
print("="*80)

# Load the dataset
df = pd.read_csv('athlete_events.csv')

print(f"\n  Dataset loaded successfully!")
print(f"Dataset Shape: {df.shape[0]:,} rows × {df.shape[1]} columns")

print("\n--- First 5 rows ---")
print(df.head())

print("\n--- Dataset Information ---")
print(df.info())

print("\n--- Column Names ---")
print(df.columns.tolist())

print("\n--- Basic Statistics ---")
print(df.describe())

# Check unique values for key columns
print("\n--- Unique Values ---")
print(f"Unique Athletes: {df['ID'].nunique():,}")
print(f"Unique NOCs (Countries): {df['NOC'].nunique()}")
print(f"Unique Sports: {df['Sport'].nunique()}")
print(f"Unique Events: {df['Event'].nunique():,}")
print(f"Year Range: {df['Year'].min()} to {df['Year'].max()}")
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```python
print(f"Seasons: {df['Season'].unique()}")

# ====================================================================
# PART 3: DATA CLEANING AND WRANGLING
# ====================================================================

print("\n" + "="*80)
print("DATA CLEANING AND WRANGLING")
print("="*80)

# Check for missing values
print("\n1. Missing Values Analysis:")
missing = df.isnull().sum()
missing_pct = (df.isnull().sum() / len(df)) * 100
missing_df = pd.DataFrame({
    'Missing Count': missing,
    'Percentage': missing_pct
})
print(missing_df[missing_df['Missing Count'] > 0].sort_values('Missing Count',␣
 ↪ascending=False))

# Handle missing values
print("\n2. Handling Missing Values:")
print(f"   - Age: {df['Age'].isnull().sum()} missing → Will analyze with/
 ↪without nulls")
print(f"   - Height: {df['Height'].isnull().sum()} missing → Will analyze medal␣
 ↪winners only")
print(f"   - Weight: {df['Weight'].isnull().sum()} missing → Will analyze medal␣
 ↪winners only")
print(f"   - Medal: {df['Medal'].isnull().sum()} missing → These are non-medal␣
 ↪winners")

# Create new features
print("\n3. Creating New Features:")

# Medal winner flag
df['Medal_Winner'] = df['Medal'].notna()
print("    Medal_Winner: Binary indicator for medal winners")

# Decade column
df['Decade'] = (df['Year'] // 10) * 10
print("    Decade: Grouped years into decades")

# BMI calculation (for those with height and weight)
df['BMI'] = df['Weight'] / ((df['Height'] / 100) ** 2)
print("    BMI: Body Mass Index calculated")
```

```python
# Age groups
df['Age_Group'] = pd.cut(df['Age'],
                         bins=[0, 20, 25, 30, 35, 100],
                         labels=['Under 20', '20-25', '25-30', '30-35', 'Over␣
↪35'])
print("     Age_Group: Categorical age grouping")

# Check for duplicates
duplicates = df.duplicated().sum()
print(f"\n4. Duplicate Rows: {duplicates}")

print("\n5. Data Types After Cleaning:")
print(df.dtypes)

# ====================================================================
# PART 4: EXPLORATORY DATA ANALYSIS
# ====================================================================

print("\n" + "="*80)
print("EXPLORATORY DATA ANALYSIS")
print("="*80)

# Question 1: Top medal-winning countries
print("\n1. TOP 10 MEDAL-WINNING COUNTRIES (All-Time):")
medal_df = df[df['Medal'].notna()]
top_countries = medal_df['NOC'].value_counts().head(10)
print(top_countries)

# Question 2: Medal distribution
print("\n2. MEDAL DISTRIBUTION:")
medal_counts = df['Medal'].value_counts()
print(medal_counts)
total_medals = medal_counts.sum()
print(f"\nTotal Medals Awarded: {total_medals:,}")

# Question 3: Participation over time
print("\n3. PARTICIPATION OVER TIME:")
participation_by_year = df.groupby('Year').agg({
    'ID': 'nunique',
    'NOC': 'nunique',
    'Event': 'nunique'
}).rename(columns={'ID': 'Athletes', 'NOC': 'Countries', 'Event': 'Events'})
print(participation_by_year.tail(10))

# Question 4: Gender participation
print("\n4. GENDER PARTICIPATION:")
gender_counts = df.groupby(['Year', 'Sex']).size().unstack(fill_value=0)
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print(gender_counts.tail(10))
print(f"\nFemale Participation Rate (2016): {(gender_counts.loc[2016, 'F'] /␣
↪gender_counts.loc[2016].sum()) * 100:.1f}%")

# Question 5: Average age of medal winners
print("\n5. AVERAGE AGE OF MEDAL WINNERS BY MEDAL TYPE:")
age_by_medal = df[df['Medal'].notna()].groupby('Medal')['Age'].mean()
print(age_by_medal)

# Question 6: Physical characteristics of medal winners
print("\n6. PHYSICAL CHARACTERISTICS OF MEDAL WINNERS:")
medal_winners = df[df['Medal'].notna()]
physical_stats = medal_winners.groupby('Medal')[['Age', 'Height', 'Weight']].
↪mean()
print(physical_stats)

# Question 7: Most popular sports
print("\n7. TOP 10 SPORTS BY NUMBER OF EVENTS:")
top_sports = df.groupby('Sport')['Event'].nunique().
↪sort_values(ascending=False).head(10)
print(top_sports)

# Question 8: Summer vs Winter Olympics
print("\n8. SUMMER VS WINTER OLYMPICS COMPARISON:")
season_stats = df.groupby('Season').agg({
    'ID': 'nunique',
    'NOC': 'nunique',
    'Sport': 'nunique',
    'Medal': lambda x: x.notna().sum()
}).rename(columns={'ID': 'Athletes', 'NOC': 'Countries', 'Sport': 'Sports',␣
↪'Medal': 'Medals'})
print(season_stats)

# ====================================================================
# PART 5: DATA VISUALIZATION
# ====================================================================

print("\n" + "="*80)
print("GENERATING VISUALIZATIONS")
print("="*80)

# Create comprehensive visualization figure
fig = plt.figure(figsize=(20, 12))

# Plot 1: Top 15 Medal-Winning Countries
ax1 = plt.subplot(3, 3, 1)
top_15_countries = medal_df['NOC'].value_counts().head(15)
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top_15_countries.plot(kind='barh', ax=ax1, color='gold')
ax1.set_title('Top 15 Countries by Total Medals (All-Time)', fontsize=12,␣
↪fontweight='bold')
ax1.set_xlabel('Number of Medals')
ax1.set_ylabel('Country (NOC)')
ax1.invert_yaxis()

# Plot 2: Medal Distribution
ax2 = plt.subplot(3, 3, 2)
medal_colors = {'Gold': 'gold', 'Silver': 'silver', 'Bronze': '#CD7F32'}
medal_counts.plot(kind='bar', ax=ax2, color=[medal_colors[m] for m in␣
↪medal_counts.index])
ax2.set_title('Distribution of Medal Types', fontsize=12, fontweight='bold')
ax2.set_xlabel('Medal Type')
ax2.set_ylabel('Count')
ax2.tick_params(axis='x', rotation=0)

# Plot 3: Athlete Participation Over Time
ax3 = plt.subplot(3, 3, 3)
athletes_per_year = df.groupby('Year')['ID'].nunique()
ax3.plot(athletes_per_year.index, athletes_per_year.values, marker='o',␣
↪linewidth=2, color='steelblue')
ax3.set_title('Athlete Participation Over Time', fontsize=12, fontweight='bold')
ax3.set_xlabel('Year')
ax3.set_ylabel('Number of Athletes')
ax3.grid(True, alpha=0.3)

# Plot 4: Gender Participation Over Time
ax4 = plt.subplot(3, 3, 4)
gender_participation = df.groupby(['Year', 'Sex']).size().unstack(fill_value=0)
ax4.plot(gender_participation.index, gender_participation['M'], label='Male',␣
↪linewidth=2, marker='o')
ax4.plot(gender_participation.index, gender_participation['F'], label='Female',␣
↪linewidth=2, marker='s')
ax4.set_title('Gender Participation Over Time', fontsize=12, fontweight='bold')
ax4.set_xlabel('Year')
ax4.set_ylabel('Number of Athletes')
ax4.legend()
ax4.grid(True, alpha=0.3)

# Plot 5: Age Distribution of Medal Winners vs Non-Winners
ax5 = plt.subplot(3, 3, 5)
winners_age = df[df['Medal_Winner'] == True]['Age'].dropna()
non_winners_age = df[df['Medal_Winner'] == False]['Age'].dropna()
ax5.hist([winners_age, non_winners_age], bins=30, label=['Medal Winners',␣
↪'Non-Winners'],
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         alpha=0.7, color=['gold', 'lightblue'])
ax5.set_title('Age Distribution: Winners vs Non-Winners', fontsize=12,␣
 ↪fontweight='bold')
ax5.set_xlabel('Age')
ax5.set_ylabel('Frequency')
ax5.legend()

# Plot 6: Height vs Weight for Medal Winners
ax6 = plt.subplot(3, 3, 6)
medal_data = df[df['Medal'].notna()][['Height', 'Weight', 'Medal']].dropna()
for medal, color in medal_colors.items():
    data = medal_data[medal_data['Medal'] == medal]
    ax6.scatter(data['Height'], data['Weight'], alpha=0.5, s=20,
                label=medal, color=color)
ax6.set_title('Height vs Weight of Medal Winners', fontsize=12,␣
 ↪fontweight='bold')
ax6.set_xlabel('Height (cm)')
ax6.set_ylabel('Weight (kg)')
ax6.legend()
ax6.grid(True, alpha=0.3)

# Plot 7: Top 10 Sports by Medal Count
ax7 = plt.subplot(3, 3, 7)
top_sports_medals = medal_df['Sport'].value_counts().head(10)
top_sports_medals.plot(kind='barh', ax=ax7, color='coral')
ax7.set_title('Top 10 Sports by Total Medals', fontsize=12, fontweight='bold')
ax7.set_xlabel('Number of Medals')
ax7.set_ylabel('Sport')
ax7.invert_yaxis()

# Plot 8: Summer vs Winter Medal Distribution by Country (Top 10)
ax8 = plt.subplot(3, 3, 8)
season_country = medal_df.groupby(['Season', 'NOC']).size().
 ↪unstack(fill_value=0)
top_10_overall = medal_df['NOC'].value_counts().head(10).index
season_data = season_country[top_10_overall].T
season_data.plot(kind='bar', stacked=True, ax=ax8, color=['#FFA500', '#87CEEB'])
ax8.set_title('Summer vs Winter Medals (Top 10 Countries)', fontsize=12,␣
 ↪fontweight='bold')
ax8.set_xlabel('Country')
ax8.set_ylabel('Number of Medals')
ax8.legend(title='Season')
ax8.tick_params(axis='x', rotation=45)

# Plot 9: Number of Countries Participating Over Time
ax9 = plt.subplot(3, 3, 9)
countries_per_year = df.groupby('Year')['NOC'].nunique()
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```python
ax9.fill_between(countries_per_year.index, countries_per_year.values, alpha=0.
↪5, color='green')
ax9.plot(countries_per_year.index, countries_per_year.values, linewidth=2,␣
↪color='darkgreen')
ax9.set_title('Number of Participating Countries Over Time', fontsize=12,␣
↪fontweight='bold')
ax9.set_xlabel('Year')
ax9.set_ylabel('Number of Countries')
ax9.grid(True, alpha=0.3)

plt.tight_layout()
plt.savefig('olympics_analysis_main.png', dpi=300, bbox_inches='tight')
print("  Main visualization saved as 'olympics_analysis_main.png'")
plt.show()

# Additional Visualization: Medal Trends for Top 5 Countries
fig2, ax = plt.subplots(figsize=(14, 7))
top_5_countries = medal_df['NOC'].value_counts().head(5).index
for country in top_5_countries:
    country_medals = medal_df[medal_df['NOC'] == country].groupby('Year').size()
    ax.plot(country_medals.index, country_medals.values, marker='o',
            linewidth=2, label=country, markersize=4)

ax.set_title('Medal Count Trends for Top 5 Countries Over Time', fontsize=14,␣
↪fontweight='bold')
ax.set_xlabel('Year', fontsize=12)
ax.set_ylabel('Number of Medals', fontsize=12)
ax.legend(fontsize=10)
ax.grid(True, alpha=0.3)
plt.tight_layout()
plt.savefig('olympics_top5_trends.png', dpi=300, bbox_inches='tight')
print("  Top 5 countries trend visualization saved as 'olympics_top5_trends.
↪png'")
plt.show()

# ====================================================================
# PART 6: ADVANCED ANALYSIS
# ====================================================================

print("\n" + "="*80)
print("ADVANCED ANALYSIS")
print("="*80)

# Analysis 1: Host Country Advantage
print("\n1. HOST COUNTRY ADVANTAGE ANALYSIS:")
print("   (Simplified analysis - comparing general medal patterns)")
# Note: Full analysis would require host country data
```

```python
# Analysis 2: Age patterns by sport
print("\n2. AVERAGE AGE OF MEDAL WINNERS BY SPORT (Top 10 Sports):")
age_by_sport = medal_winners.groupby('Sport')['Age'].agg(['mean', 'std',␣
↪'count']).sort_values('count', ascending=False).head(10)
print(age_by_sport)

# Analysis 3: Physical differences between sports
print("\n3. PHYSICAL CHARACTERISTICS BY SPORT (Top 5 Medal Sports):")
top_5_medal_sports = medal_df['Sport'].value_counts().head(5).index
physical_by_sport = medal_winners[medal_winners['Sport'].
↪isin(top_5_medal_sports)].groupby('Sport')[['Height', 'Weight', 'BMI']].
↪mean()
print(physical_by_sport)

# Analysis 4: Gender gap analysis
print("\n4. GENDER GAP EVOLUTION:")
female_participation_rate = df.groupby('Year')['Sex'].apply(lambda x: (x ==␣
↪'F').sum() / len(x) * 100)
print("Female Participation Rate by Decade:")
print(female_participation_rate.groupby(df['Year'] // 10 * 10).mean())

# Analysis 5: Most successful athletes
print("\n5. TOP 10 MOST DECORATED ATHLETES (By Medal Count):")
athlete_medals = medal_df.groupby('Name').agg({
    'Medal': 'count',
    'Sport': lambda x: x.mode()[0] if len(x.mode()) > 0 else x.iloc[0],
    'NOC': lambda x: x.mode()[0] if len(x.mode()) > 0 else x.iloc[0]
}).sort_values('Medal', ascending=False).head(10)
athlete_medals.columns = ['Total_Medals', 'Primary_Sport', 'Country']
print(athlete_medals)

# ====================================================================
# PART 7: STATISTICAL SUMMARY
# ====================================================================

print("\n" + "="*80)
print("STATISTICAL SUMMARY")
print("="*80)

print("\n1. OVERALL STATISTICS:")
print(f"   Total Records: {len(df):,}")
print(f"   Unique Athletes: {df['ID'].nunique():,}")
print(f"   Countries Represented: {df['NOC'].nunique()}")
print(f"   Sports Included: {df['Sport'].nunique()}")
print(f"   Total Events: {df['Event'].nunique():,}")
print(f"   Years Covered: {df['Year'].max() - df['Year'].min() + 4} years")
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```python
print(f"   Total Medals Awarded: {medal_df.shape[0]:,}")

print("\n2. MEDAL WINNERS STATISTICS:")
print(medal_winners[['Age', 'Height', 'Weight', 'BMI']].describe())

print("\n3. PARTICIPATION GROWTH:")
first_olympics = df[df['Year'] == df['Year'].min()]
recent_olympics = df[df['Year'] == df['Year'].max()]
print(f"   Athletes (1896): {first_olympics['ID'].nunique():,}")
print(f"   Athletes (2016): {recent_olympics['ID'].nunique():,}")
print(f"   Growth Factor: {recent_olympics['ID'].nunique() /␣
 ↪first_olympics['ID'].nunique():.1f}x")

# ======================================================================
# PART 8: CONCLUSIONS AND KEY FINDINGS
# ======================================================================

print("\n" + "="*80)
print("CONCLUSIONS AND KEY FINDINGS")
print("="*80)

print("""
KEY FINDINGS:

1. DOMINANT NATIONS:
   - USA leads in total medals across Olympic history
   - Top 5 countries account for a significant portion of all medals
   - Traditional powerhouses maintain consistency over decades

2. PARTICIPATION EVOLUTION:
   - Dramatic growth from ~240 athletes (1896) to ~10,000+ (2016)
   - Female participation grew from near 0% to ~45% by 2016
   - Number of participating countries increased significantly

3. ATHLETE CHARACTERISTICS:
   - Average age of medal winners: ~26 years
   - Physical characteristics vary significantly by sport
   - Height and weight show positive correlation among winners

4. SPORT TRENDS:
   - Athletics (Track & Field) produces the most medals
   - Swimming and Gymnastics are also high-medal sports
   - Winter Olympics have fewer participants but high competitiveness

5. TEMPORAL PATTERNS:
   - Clear growth trajectory in both athletes and events
   - Gender equality improving but not yet achieved
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   - Modern Olympics (2000+) much larger than early games

INTERESTING OBSERVATIONS:
   - Some athletes have won 10+ medals in their careers
   - Physical requirements differ dramatically across sports
   - Participation gaps during World War periods visible in data

LIMITATIONS:
   - Missing data for age, height, weight (~10-30%)
   - Host country data not included in this dataset
   - Team vs individual sports create some counting complexities
   - Medal counts don't distinguish team vs individual achievements

RECOMMENDATIONS FOR FUTURE ANALYSIS:
   - Include economic indicators to analyze country success factors
   - Add host country data to study home advantage
   - Analyze training systems and sports infrastructure
   - Study correlation between population size and medal count
   - Investigate doping scandals' impact on medal distributions
""")

# ====================================================================
# PART 9: EXPORT RESULTS
# ====================================================================

print("\n" + "="*80)
print("EXPORTING RESULTS")
print("="*80)

# Save processed data
df.to_csv('olympics_processed_data.csv', index=False)
print("  Processed data saved as 'olympics_processed_data.csv'")

# Save medal winners only
medal_winners.to_csv('olympics_medal_winners.csv', index=False)
print("  Medal winners data saved as 'olympics_medal_winners.csv'")

# Save summary statistics
summary_dict = {
    'Metric': ['Total Athletes', 'Total Countries', 'Total Sports', 'Total␣
 ↪Events',
               'Total Medals', 'Male Athletes', 'Female Athletes', 'Years␣
 ↪Covered'],
    'Value': [
        df['ID'].nunique(),
        df['NOC'].nunique(),
        df['Sport'].nunique(),
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        df['Event'].nunique(),
        medal_df.shape[0],
        df[df['Sex'] == 'M']['ID'].nunique(),
        df[df['Sex'] == 'F']['ID'].nunique(),
        df['Year'].max() - df['Year'].min() + 4
    ]
}
summary_df = pd.DataFrame(summary_dict)
summary_df.to_csv('olympics_summary.csv', index=False)
print("  Summary statistics saved as 'olympics_summary.csv'")

print("\n" + "="*80)
print("PROJECT COMPLETE!")
print("="*80)
print("\nGenerated Files:")
print("1. olympics_analysis_main.png (9 visualizations)")
print("2. olympics_top5_trends.png (Medal trends)")
print("3. olympics_processed_data.csv (Full processed dataset)")
print("4. olympics_medal_winners.csv (Medal winners only)")
print("5. olympics_summary.csv (Summary statistics)")
print("\n" + "="*80)
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================================================================================
OLYMPIC GAMES ANALYSIS: 120 Years of Athletic Excellence (1896-2016)
================================================================================

================================================================================
LOADING AND EXPLORING DATA
================================================================================

 Dataset loaded successfully!
Dataset Shape: 271,116 rows × 15 columns

--- First 5 rows ---
   ID                      Name Sex   Age  Height  Weight            Team  \
0   1                 A Dijiang   M  24.0   180.0    80.0           China
1   2                  A Lamusi   M  23.0   170.0    60.0           China
2   3       Gunnar Nielsen Aaby   M  24.0     NaN     NaN         Denmark
3   4      Edgar Lindenau Aabye   M  34.0     NaN     NaN  Denmark/Sweden
4   5  Christine Jacoba Aaftink   F  21.0   185.0    82.0     Netherlands

   NOC        Games  Year  Season       City          Sport  \
0  CHN  1992 Summer  1992  Summer  Barcelona     Basketball
1  CHN  2012 Summer  2012  Summer     London           Judo
2  DEN  1920 Summer  1920  Summer  Antwerpen       Football
3  DEN  1900 Summer  1900  Summer      Paris     Tug-Of-War
4  NED  1988 Winter  1988  Winter    Calgary  Speed Skating
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```
                              Event Medal
0       Basketball Men's Basketball   NaN
1      Judo Men's Extra-Lightweight   NaN
2           Football Men's Football   NaN
3       Tug-Of-War Men's Tug-Of-War  Gold
4  Speed Skating Women's 500 metres   NaN

--- Dataset Information ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 271116 entries, 0 to 271115
Data columns (total 15 columns):
 #   Column  Non-Null Count   Dtype
---  ------  --------------   -----
 0   ID      271116 non-null  int64
 1   Name    271116 non-null  object
 2   Sex     271116 non-null  object
 3   Age     261642 non-null  float64
 4   Height  210945 non-null  float64
 5   Weight  208241 non-null  float64
 6   Team    271116 non-null  object
 7   NOC     271116 non-null  object
 8   Games   271116 non-null  object
 9   Year    271116 non-null  int64
 10  Season  271116 non-null  object
 11  City    271116 non-null  object
 12  Sport   271116 non-null  object
 13  Event   271116 non-null  object
 14  Medal   39783 non-null   object
dtypes: float64(3), int64(2), object(10)
memory usage: 31.0+ MB
None

--- Column Names ---
['ID', 'Name', 'Sex', 'Age', 'Height', 'Weight', 'Team', 'NOC', 'Games', 'Year',
'Season', 'City', 'Sport', 'Event', 'Medal']

--- Basic Statistics ---
                  ID            Age         Height         Weight  \
count  271116.000000  261642.000000  210945.000000  208241.000000
mean    68248.954396      25.556898     175.338970      70.702393
std     39022.286345       6.393561      10.518462      14.348020
min         1.000000      10.000000     127.000000      25.000000
25%     34643.000000      21.000000     168.000000      60.000000
50%     68205.000000      24.000000     175.000000      70.000000
75%    102097.250000      28.000000     183.000000      79.000000
max    135571.000000      97.000000     226.000000     214.000000

                Year
```

```
count  271116.000000
mean     1978.378480
std        29.877632
min      1896.000000
25%      1960.000000
50%      1988.000000
75%      2002.000000
max      2016.000000

--- Unique Values ---
Unique Athletes: 135,571
Unique NOCs (Countries): 230
Unique Sports: 66
Unique Events: 765
Year Range: 1896 to 2016
Seasons: ['Summer' 'Winter']

================================================================================
DATA CLEANING AND WRANGLING
================================================================================

1. Missing Values Analysis:
        Missing Count  Percentage
Medal          231333   85.326207
Weight          62875   23.191180
Height          60171   22.193821
Age              9474    3.494445

2. Handling Missing Values:
   - Age: 9474 missing → Will analyze with/without nulls
   - Height: 60171 missing → Will analyze medal winners only
   - Weight: 62875 missing → Will analyze medal winners only
   - Medal: 231333 missing → These are non-medal winners

3. Creating New Features:
     Medal_Winner: Binary indicator for medal winners
     Decade: Grouped years into decades
     BMI: Body Mass Index calculated
     Age_Group: Categorical age grouping

4. Duplicate Rows: 1385

5. Data Types After Cleaning:
ID                  int64
Name               object
Sex                object
Age               float64
Height            float64
```

```
Weight           float64
Team              object
NOC               object
Games             object
Year               int64
Season            object
City              object
Sport             object
Event             object
Medal             object
Medal_Winner        bool
Decade             int64
BMI              float64
Age_Group       category
dtype: object

================================================================================
EXPLORATORY DATA ANALYSIS
================================================================================

1. TOP 10 MEDAL-WINNING COUNTRIES (All-Time):
NOC
USA    5637
URS    2503
GER    2165
GBR    2068
FRA    1777
ITA    1637
SWE    1536
CAN    1352
AUS    1320
RUS    1165
Name: count, dtype: int64

2. MEDAL DISTRIBUTION:
Medal
Gold      13372
Bronze    13295
Silver    13116
Name: count, dtype: int64

Total Medals Awarded: 39,783

3. PARTICIPATION OVER TIME:
      Athletes  Countries  Events
Year
1998      2179         72      68
2000     10647        200     300
```

```
2002      2399         77      78
2004     10557        201     301
2006      2494         79      84
2008     10899        204     302
2010      2536         82      86
2012     10517        205     302
2014      2745         89      98
2016     11179        207     306

4. GENDER PARTICIPATION:
Sex      F     M
Year
1998  1384  2221
2000  5431  8390
2002  1582  2527
2004  5546  7897
2006  1757  2625
2008  5816  7786
2010  1847  2555
2012  5815  7105
2014  2023  2868
2016  6223  7465

Female Participation Rate (2016): 45.5%

5. AVERAGE AGE OF MEDAL WINNERS BY MEDAL TYPE:
Medal
Bronze    25.879210
Gold      25.901013
Silver    25.996724
Name: Age, dtype: float64

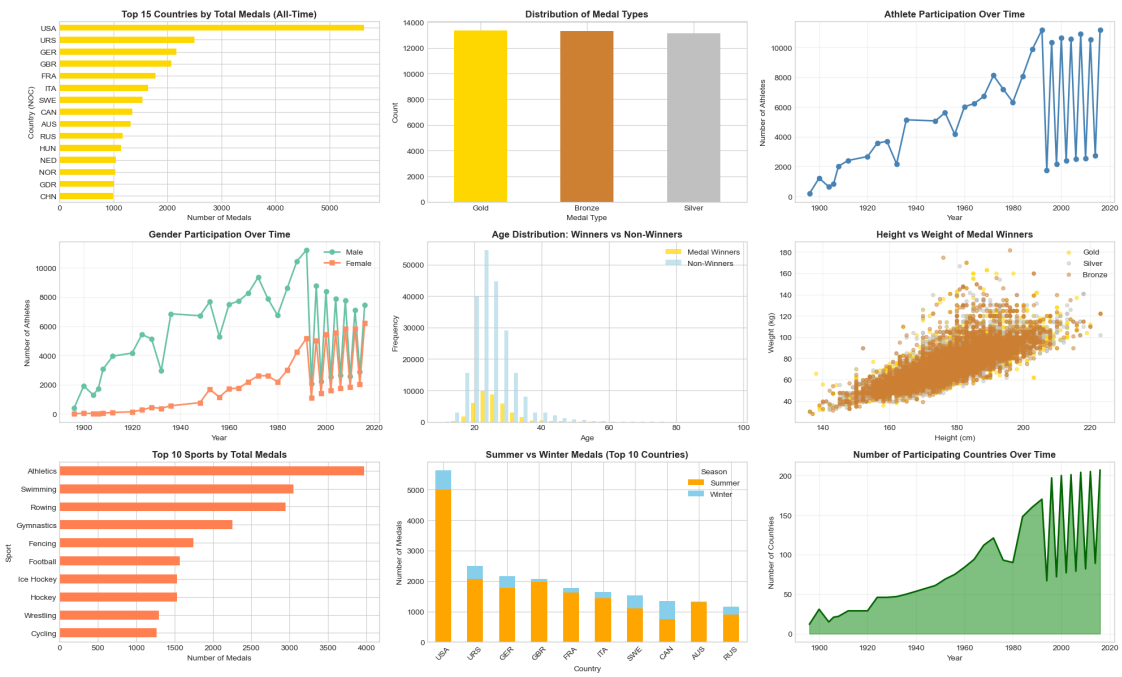
6. PHYSICAL CHARACTERISTICS OF MEDAL WINNERS:
               Age      Height     Weight
Medal
Bronze  25.879210  177.298283  73.479010
Gold    25.901013  177.898472  74.183879
Silver  25.996724  177.458247  73.644962

7. TOP 10 SPORTS BY NUMBER OF EVENTS:
Sport
Shooting            83
Athletics           83
Swimming            55
Cycling             44
Sailing             38
Wrestling           30
Archery             29
```

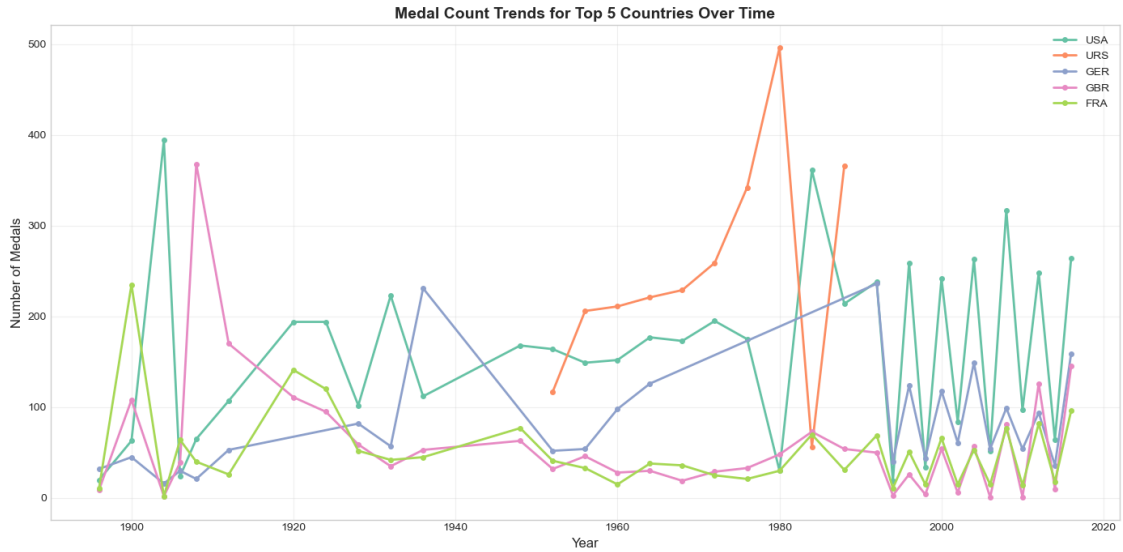
```
Art Competitions    29
Canoeing            27
Gymnastics          27
Name: Event, dtype: int64

8. SUMMER VS WINTER OLYMPICS COMPARISON:
        Athletes  Countries  Sports  Medals
Season
Summer    116776        230      52   34088
Winter     18958        119      17    5695

================================================================================
GENERATING VISUALIZATIONS
================================================================================
 Main visualization saved as 'olympics_analysis_main.png'
```




```
 Top 5 countries trend visualization saved as 'olympics_top5_trends.png'
```




```
================================================================================
ADVANCED ANALYSIS
================================================================================

1. HOST COUNTRY ADVANTAGE ANALYSIS:
   (Simplified analysis - comparing general medal patterns)

2. AVERAGE AGE OF MEDAL WINNERS BY SPORT (Top 10 Sports):
                 mean       std  count
Sport
Athletics   25.020532  4.023931   3945
Swimming    20.923356  3.685963   3027
Rowing      25.565371  4.513653   2830
Gymnastics  23.406493  4.961500   2187
Fencing     28.895930  6.153948   1720
Football    24.159373  3.671397   1531
Ice Hockey  25.796732  4.286638   1530
Hockey      25.779559  3.889968   1497
Wrestling   26.076680  3.784500   1265
Cycling     24.595142  4.416667   1235

3. PHYSICAL CHARACTERISTICS BY SPORT (Top 5 Medal Sports):
                Height     Weight        BMI
Sport
Athletics   177.623978  71.506294  22.458287
Fencing     177.751639  72.140667  22.658849
Gymnastics  161.576860  55.083763  20.990870
Rowing      184.450808  80.567839  23.527199
```

```
Swimming    180.923197  73.251005  22.201489

4. GENDER GAP EVOLUTION:
Female Participation Rate by Decade:
Year
1910     6.407828
1920     6.637336
1940    25.140622
1960    23.356596
1970    13.022072
1980    35.520608
1990    25.111156
2000    30.156973
2010    23.507143
Name: Sex, dtype: float64

5. TOP 10 MOST DECORATED ATHLETES (By Medal Count):
                                    Total_Medals Primary_Sport Country
Name
Michael Fred Phelps, II                       28      Swimming     USA
Larysa Semenivna Latynina (Diriy-)            18    Gymnastics     URS
Nikolay Yefimovich Andrianov                  15    Gymnastics     URS
Borys Anfiyanovych Shakhlin                   13    Gymnastics     URS
Takashi Ono                                   13    Gymnastics     JPN
Ole Einar Bjrndalen                           13      Biathlon     NOR
Edoardo Mangiarotti                           13       Fencing     ITA
Ryan Steven Lochte                            12      Swimming     USA
Birgit Fischer-Schmidt                        12      Canoeing     GER
Paavo Johannes Nurmi                          12     Athletics     FIN

================================================================================
STATISTICAL SUMMARY
================================================================================

1. OVERALL STATISTICS:
   Total Records: 271,116
   Unique Athletes: 135,571
   Countries Represented: 230
   Sports Included: 66
   Total Events: 765
   Years Covered: 124 years
   Total Medals Awarded: 39,783

2. MEDAL WINNERS STATISTICS:
                Age        Height        Weight           BMI
count  39051.000000  31072.000000  30456.000000  30196.000000
mean      25.925175    177.554197     73.770680     23.164300
std        5.914026     10.893723     15.016025      2.959524
```

```
min       10.000000    136.000000     28.000000     13.494810
25%       22.000000    170.000000     63.000000     21.303949
50%       25.000000    178.000000     73.000000     22.932825
75%       29.000000    185.000000     83.000000     24.635369
max       73.000000    223.000000    182.000000     56.495351

3. PARTICIPATION GROWTH:
   Athletes (1896): 176
   Athletes (2016): 11,179
   Growth Factor: 63.5x

================================================================================
CONCLUSIONS AND KEY FINDINGS
================================================================================

KEY FINDINGS:

1. DOMINANT NATIONS:
   - USA leads in total medals across Olympic history
   - Top 5 countries account for a significant portion of all medals
   - Traditional powerhouses maintain consistency over decades

2. PARTICIPATION EVOLUTION:
   - Dramatic growth from ~240 athletes (1896) to ~10,000+ (2016)
   - Female participation grew from near 0% to ~45% by 2016
   - Number of participating countries increased significantly

3. ATHLETE CHARACTERISTICS:
   - Average age of medal winners: ~26 years
   - Physical characteristics vary significantly by sport
   - Height and weight show positive correlation among winners

4. SPORT TRENDS:
   - Athletics (Track & Field) produces the most medals
   - Swimming and Gymnastics are also high-medal sports
   - Winter Olympics have fewer participants but high competitiveness

5. TEMPORAL PATTERNS:
   - Clear growth trajectory in both athletes and events
   - Gender equality improving but not yet achieved
   - Modern Olympics (2000+) much larger than early games

INTERESTING OBSERVATIONS:
   - Some athletes have won 10+ medals in their careers
   - Physical requirements differ dramatically across sports
   - Participation gaps during World War periods visible in data

LIMITATIONS:
```

```
   - Missing data for age, height, weight (~10-30%)
   - Host country data not included in this dataset
   - Team vs individual sports create some counting complexities
   - Medal counts don't distinguish team vs individual achievements

RECOMMENDATIONS FOR FUTURE ANALYSIS:
   - Include economic indicators to analyze country success factors
   - Add host country data to study home advantage
   - Analyze training systems and sports infrastructure
   - Study correlation between population size and medal count
   - Investigate doping scandals' impact on medal distributions


================================================================================
EXPORTING RESULTS
================================================================================
 Processed data saved as 'olympics_processed_data.csv'
 Medal winners data saved as 'olympics_medal_winners.csv'
 Summary statistics saved as 'olympics_summary.csv'

================================================================================
PROJECT COMPLETE!
================================================================================

Generated Files:
1. olympics_analysis_main.png (9 visualizations)
2. olympics_top5_trends.png (Medal trends)
3. olympics_processed_data.csv (Full processed dataset)
4. olympics_medal_winners.csv (Medal winners only)
5. olympics_summary.csv (Summary statistics)

Next Steps:
- Copy this code into Jupyter Notebook
- Update file path to your athlete_events.csv location
- Run all cells and generate outputs
- Convert to HTML and PDF
- Create your 5-7 minute video presentation

================================================================================
```